

Intelligent **SDN** Security

DDoS Detection and Mitigation using LSTM Deep Learning on CICIDS2019
Dataset

Project **Architecture**

Combining AI intelligence with Software Defined Networking for proactive defense.

CICIDS2019 **Dataset** Analysis

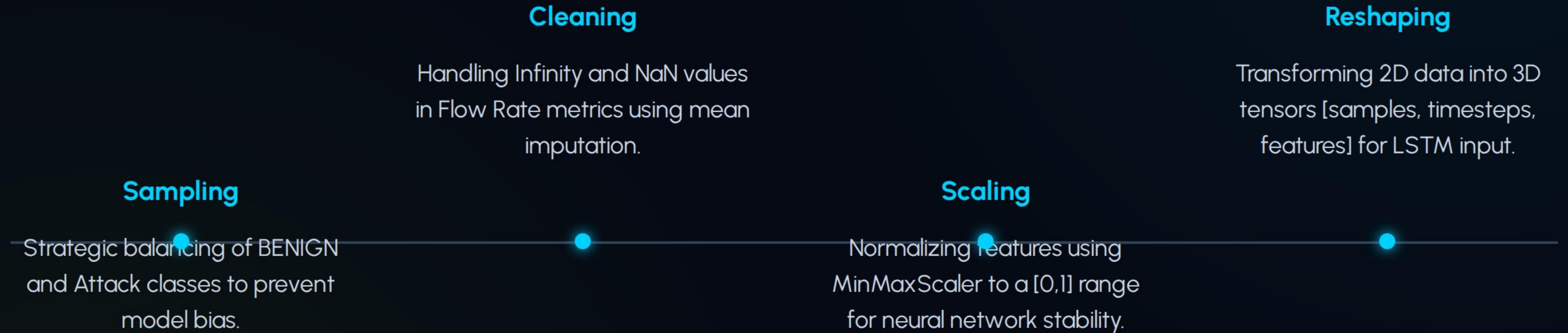
Modern Attack Vectors

Features 2019's most prevalent DDoS attacks including Reflection (NTP, DNS, LDAP) and Exploitation (SYN Flood, UDP Flood).

Rich Feature Space

Contains over 80 network flow features extracted via CICFlowMeter, providing deep temporal and statistical insights for ML models.

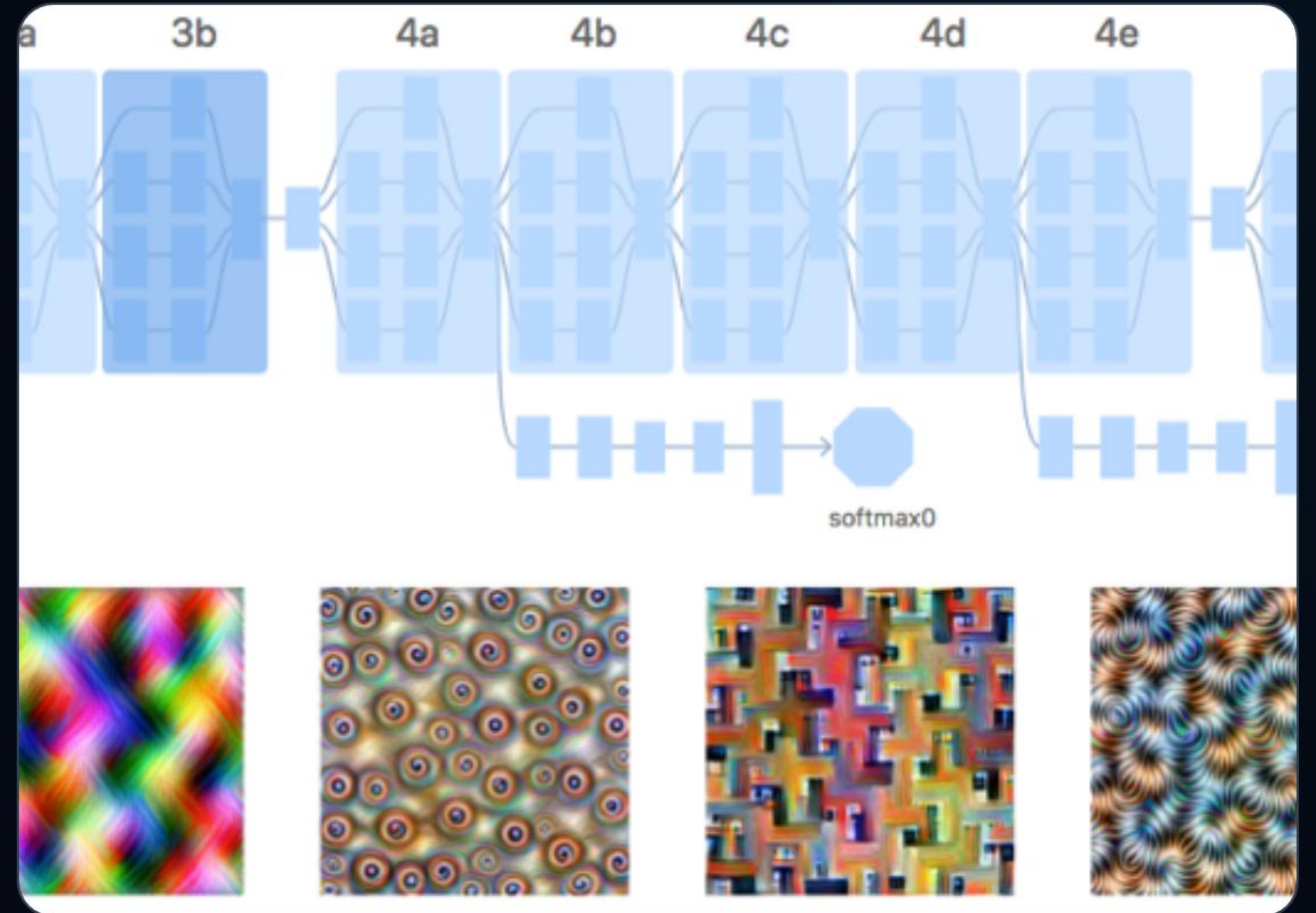
Data Preprocessing Pipeline



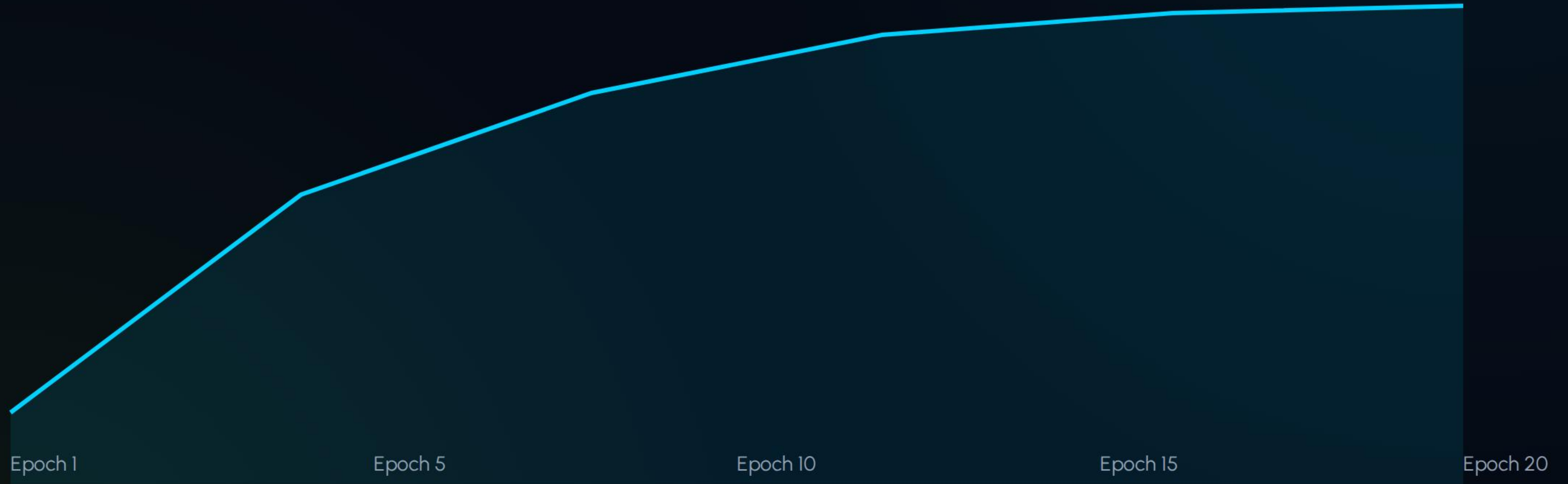
LSTM **Model** Architecture

Sequential Memory Layers

- **Layer 1:** 64 LSTM units capturing temporal dependencies.
- **Layer 2:** 32 LSTM units for hierarchical feature extraction.
- **Regularization:** 20% Dropout layers to prevent overfitting.
- **Optimization:** Adam optimizer with Sparse Categorical Crossentropy.



Training **Dynamics** Analysis



Loss convergence over 20 epochs shows rapid initial learning and stability (Good Fit).

Why **LSTM?** Comparative View

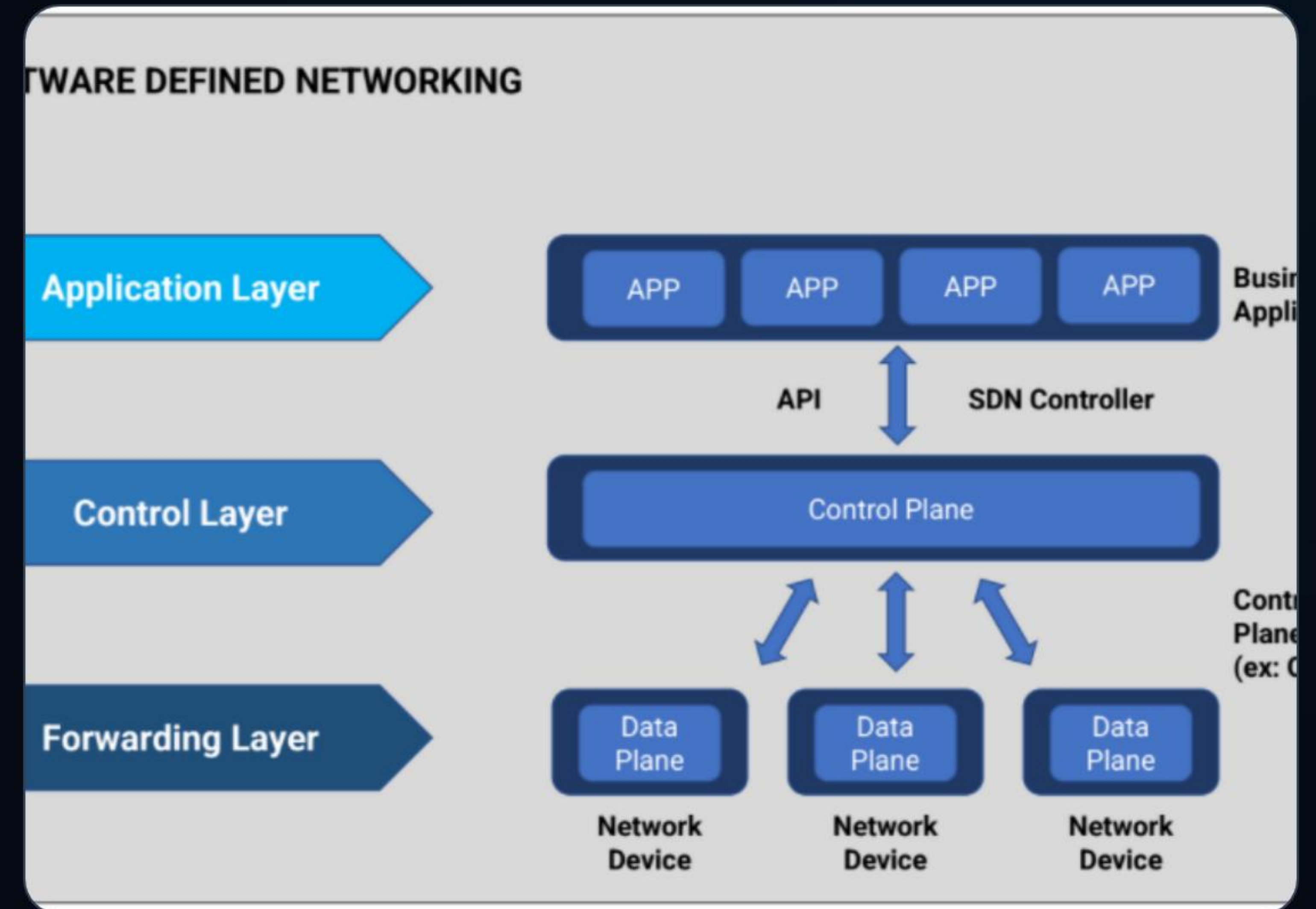
Feature	Decision Tree	Random Forest	LSTM (Proposed)
Data Logic	Tabular / IF-ELSE	Ensemble Voting	Temporal / Sequential
Feature Extraction	Manual Selection	Weighted Importance	Auto-Learned Context
Pattern Memory	Low	Moderate	High (Memory Cells)
SDN Scalability	Static Thresholds	Resource Heavy	Real-time Windowing

Ryu **Controller** Integration

The Intelligence Loop

The Ryu controller periodically requests Port Statistics (OpenFlow PortStatsReply). These flow-level metrics are fed into the LSTM agent.

- Real-time flow inspection (PPS/BPS).
- Asynchronous AI inference.
- Automated Flow-Mod rule injection.



Intelligent **Mitigation** Engine



Detection

AI monitors traffic in 2-second intervals, checking against a 0.90 confidence threshold.



Verification

Sliding window logic (CONSECUTIVE_HITS) ensures defense is only triggered by persistent attacks.



Mitigation

Automated Flow Modification (Flow-Mod) drops packets from the malicious source IP for 90 seconds.

Real-time Dashboard

Visual Analytics

The system provides a real-time monitoring interface that displays traffic rates (PPS/BPS) per port and AI prediction labels.

Administrators can track the "Confidence Score" and "Model ID" currently active in the SDN ecosystem.



Performance **Benchmarks**



The LSTM model achieves high-precision detection with minimal inference latency suitable for SDN controllers.

Questions?

Thank you for your attention.

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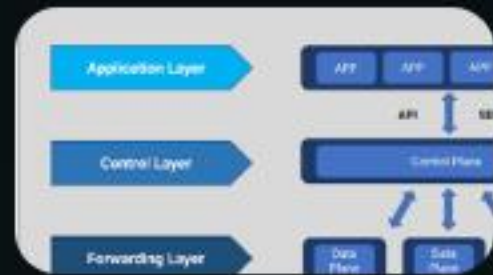
Image Sources



Thumbnail
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